PART A — MODULE 1 QUESTIONS

MATH11-M1-001
Prompt:
What is the solution to $11 - (2x)/3 = 5$?
A) 6
B) 8
C) 9
D) 12

MATH11-M1-002
Prompt:
A student completes 42 practice questions in 6 days at a constant rate. At this rate, how many practice questions does the student complete in 9 days?
A) 54
B) 60
C) 63
D) 72

MATH11-M1-003
Prompt:
The table shows the number of books checked out by students during one week.

Books checked out: 0, 1, 2, 3
Number of students: 4, 7, 6, 5

How many students checked out exactly 2 books?
A) 5
B) 6
C) 7
D) 12

MATH11-M1-004
Prompt:
A linear function f has a rate of change of 5. If $f(3) = 2$, what is $f(0)$?
A) −15
B) −13
C) −3
D) 17

MATH11-M1-005
Prompt:
A student has at most \$40 to spend on a school project. The student spends \$12 on supplies and then buys m sheets of poster board for \$2.50 each. Which inequality represents this situation?
A) $12 + 2.50m \leq 40$
B) $12 + 2.50m \geq 40$
C) $2.50 + 12m \leq 40$
D) $2.50m - 12 \leq 40$

MATH11-M1-006
Prompt:

What is the value of y in the solution to the system of equations?
$x + 3y = 19$
$x = 7$
A) 3
B) 4
C) 6
D) 12

MATH11-M1-007
Prompt:
Which expression is equivalent to $5(3t + 2) - 2(t - 6)$?
A) $13t - 2$
B) $13t + 22$
C) $17t - 2$
D) $17t + 22$

MATH11-M1-008
Prompt:
Which expression is equivalent to $(r^6 \div r^2)(r^3)$, where $r \neq 0$?
A) r^5
B) r^7
C) r^8
D) r^{11}

MATH11-M1-009
Prompt:
A box contains 8 green clips, 10 silver clips, and 6 gold clips. If one clip is selected at random, what is the probability that the clip selected is silver?
A) 1/4
B) 1/3
C) 5/12
D) 3/4

MATH11-M1-010
Prompt:
An equilateral triangle has perimeter 42 inches. What is the length, in inches, of one side of the triangle?
A) 12
B) 14
C) 21
D) 39

MATH11-M1-011
Prompt:
In the xy-plane, point P has coordinates $(-5, 2)$. Point Q is 3 units to the right and 4 units down from point P. What are the coordinates of point Q?
A) $(-8, 6)$
B) $(-8, -2)$
C) $(-2, 6)$
D) $(-2, -2)$

MATH11-M1-012
Prompt:
What is the value of $\sqrt{225} \div 5$?
A) 3
B) 5
C) 15
D) 45

MATH11-M1-013
Prompt:
If $h(x) = x^2 - 4x + 1$, what is $h(-1)$?
A) -4
B) -2
C) 6
D) 10

MATH11-M1-014
Prompt:
The equation $(x - 5)(x + 1) = 0$ has two solutions. What is the greater solution?
A) -5
B) -1
C) 1
D) 5

MATH11-M1-015
Prompt:
A store had 120 cans of soup on a shelf. By the end of the day, 30 cans had been sold. What fraction of the cans had been sold?
A) 1/5
B) 1/4
C) 1/3
D) 3/4

MATH11-M1-016
Prompt:
The table shows a linear relationship between x and y.

x: 1, 3, 5, 7
y: 9, 15, 21, 27

Which equation represents this relationship?
A) $y = 2x + 7$
B) $y = 3x + 6$
C) $y = 4x + 5$
D) $y = 6x + 3$

MATH11-M1-017
Prompt:
A rectangle has length 20 centimeters and diagonal length 25 centimeters. What is the width, in centimeters, of the rectangle?
A) 10

B) 12
C) 15
D) 18

MATH11-M1-018
Prompt:
The formula $A = \pi r^2$ gives the area A of a circle with radius r. What is the area of a circle with radius 8 meters?
A) 16π
B) 32π
C) 64π
D) 128π

MATH11-M1-019
Prompt:
The equation $C = 45 + 6h$ models the total cost C, in dollars, to rent a room for h hours. What does 45 represent in this model?
A) The cost per hour, in dollars
B) The fixed cost, in dollars, before any hourly charges
C) The number of hours the room is rented
D) The total cost, in dollars, for 45 hours

MATH11-M1-020
Prompt:
A box contains 25 pencils. Of the pencils, 15 are sharpened. What percent of the pencils are sharpened?
A) 15%
B) 25%
C) 40%
D) 60%

MATH11-M1-021
Prompt:
The values in a data set are 5, 9, 9, 12, 15, and 16. What is the mode of the data set?
A) 9
B) 11
C) 12
D) 15

MATH11-M1-022
Prompt:
A cylinder has radius 3 inches and height 5 inches. The volume of a cylinder is $V = \pi r^2 h$. What is the volume, in cubic inches, of the cylinder?
A) 15π
B) 30π
C) 45π
D) 90π

PART B — MODULE 2 QUESTIONS

MATH11-M2-001
Prompt:
What is the solution to $3(2x - 5) + 4 = 5x + 11$?
A) 12
B) 18
C) 20
D) 22

MATH11-M2-002
Prompt:
The system of equations has solution (x, y).
$3x + y = 31$
$2x - 3y = -3$
What is the value of $x + y$?
A) 9
B) 11
C) 13
D) 15

MATH11-M2-003
Prompt:
The function f is defined by $f(x) = 3x^2 - 18x + 20$. What is the minimum value of $f(x)$?
A) -7
B) -3
C) 3
D) 20

MATH11-M2-004
Prompt:
For $x \neq -2$, which expression is equivalent to $(x^2 - 3x - 10)/(x + 2)$?
A) $x - 5$
B) $x - 2$
C) $x + 3$
D) $x + 5$

MATH11-M2-005
Prompt:
A quantity is modeled by the expression $640(0.5)^t$, where t is the number of years after the quantity was first measured. What does the factor 0.5 indicate?
A) The quantity increases by 50% each year.
B) The quantity decreases by 50% each year.
C) The quantity starts at 0.5.
D) The quantity is 640 after 0.5 year.

MATH11-M2-006
Prompt:
A gym increased its monthly membership fee by 25%. After the increase, the monthly fee was $75.
What was the monthly fee before the increase?
A) $50
B) $55

C) $60
D) $65

MATH11-M2-007
Prompt:
The table shows the number of students in a program by grade and by whether they studied a world language.

Grade: 9, 10, 11
Studied a world language: 14, 18, 16
Did not study a world language: 6, 12, 14

If one student from the program is selected at random, what is the probability that the student studied a world language?
A) 3/8
B) 1/2
C) 3/5
D) 4/5

MATH11-M2-008
Prompt:
A rectangular swimming pool is 18 meters long and 10 meters wide. A walkway of uniform width 2 meters surrounds the pool. What is the area, in square meters, of the walkway?
A) 64
B) 96
C) 128
D) 180

MATH11-M2-009
Prompt:
In a right triangle, the side opposite angle θ has length 20 and the side adjacent to angle θ has length 21. What is $\sin \theta$?
A) 20/29
B) 20/21
C) 21/29
D) 29/20

MATH11-M2-010
Prompt:
A line passes through the points $(1, -7)$ and $(9, 5)$. What is the x-intercept of the line?
A) 8/3
B) 11/3
C) 14/3
D) 17/3

MATH11-M2-011
Prompt:
The functions f and g are defined by $f(x) = 5x - 2$ and $g(x) = x^2 + x$. What is $f(g(3))$?
A) 48
B) 58

C) 60
D) 62

MATH11-M2-012
Prompt:
What is the solution to $\sqrt{(4x + 9)} = x$?
A) 1
B) 3
C) 5
D) 9

MATH11-M2-013
Prompt:
In the xy-plane, a circle has equation $(x - 6)^2 + (y + 2)^2 = 49$. What is the center of the circle?
A) $(-6, 2)$
B) $(-6, -2)$
C) $(6, -2)$
D) $(6, 2)$

MATH11-M2-014
Prompt:
For what value of a does the equation $ax + 9 = 4x + 9$ have infinitely many solutions?
A) 0
B) 4
C) 5
D) 9

MATH11-M2-015
Prompt:
The table shows values of a quadratic function p.

x: $-1, 0, 1, 2$
p(x): $-4, 1, 2, -1$

Which equation could define p?
A) $p(x) = -2x^2 + 3x + 1$
B) $p(x) = -2x^2 - 3x + 1$
C) $p(x) = 2x^2 + 3x - 1$
D) $p(x) = 2x^2 - 3x - 1$

MATH11-M2-016
Prompt:
Which expression is equivalent to $(12x^4y^3 + 18x^2y^2)/(6x^2y^2)$, where $x \neq 0$ and $y \neq 0$?
A) $2x^2y + 3$
B) $2x^2y + 18$
C) $6x^2y + 3$
D) $12x^2y + 18$

MATH11-M2-017
Prompt:
In a survey of 120 students, 70 students said they use a tablet for schoolwork, 58 said they use a

laptop for schoolwork, and 32 said they use both a tablet and a laptop. How many students said they use exactly one of these two devices for schoolwork?
A) 64
B) 70
C) 96
D) 108

MATH11-M2-018
Prompt:
Two similar prisms have corresponding side lengths in the ratio 3:4. The surface area of the smaller prism is 135 square centimeters. What is the surface area, in square centimeters, of the larger prism?
A) 180
B) 210
C) 240
D) 320

MATH11-M2-019
Prompt:
For all values of x, $2(x + 3)^2 - k = 2x^2 + 12x + 11$. What is the value of k?
A) 7
B) 11
C) 18
D) 29

MATH11-M2-020
Prompt:
A data set has 5 values with a mean of 32. If one value, 44, is removed, what is the mean of the remaining 4 values?
A) 27
B) 29
C) 30
D) 31

MATH11-M2-021
Prompt:
At a school event, wristbands cost $3 each and meal tickets cost $8 each. A total of 52 wristbands and meal tickets were sold for $256. How many meal tickets were sold?
A) 16
B) 18
C) 20
D) 22

MATH11-M2-022
Prompt:
The graph of a quadratic function has vertex (2, −6) and passes through the point (5, 12). Which equation could define the function?
A) $y = 2(x - 2)^2 - 6$
B) $y = 2(x + 2)^2 - 6$
C) $y = -2(x - 2)^2 - 6$
D) $y = (x - 2)^2 + 12$

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH11-M1-001 — Correct: 9 — Explanation: Subtract 11 from both sides to get $-(2x)/3 = -6$. Multiply by $-3/2$ to get $x = 9$.
MATH11-M1-002 — Correct: 63 — Explanation: The student completes $42 \div 6 = 7$ questions per day. In 9 days, the student completes $7 \times 9 = 63$ questions.
MATH11-M1-003 — Correct: 6 — Explanation: The table shows that 6 students checked out exactly 2 books.
MATH11-M1-004 — Correct: −13 — Explanation: A rate of change of 5 means f increases by 5 for each increase of 1 in x. From $x = 0$ to $x = 3$, the increase is 15, so $f(0) = 2 - 15 = -13$.
MATH11-M1-005 — Correct: $12 + 2.50m \leq 40$ — Explanation: The total cost is the $12 supply cost plus $2.50 for each of m poster boards, and this total is at most $40.
MATH11-M1-006 — Correct: 4 — Explanation: Substitute $x = 7$ into $x + 3y = 19$ to get $7 + 3y = 19$. Then $3y = 12$, so $y = 4$.
MATH11-M1-007 — Correct: $13t + 22$ — Explanation: Distribute and combine like terms: $5(3t + 2) - 2(t - 6) = 15t + 10 - 2t + 12 = 13t + 22$.
MATH11-M1-008 — Correct: r^7 — Explanation: First, $r^6 \div r^2 = r^4$. Then $r^4 \times r^3 = r^7$.
MATH11-M1-009 — Correct: 5/12 — Explanation: There are $8 + 10 + 6 = 24$ clips total, and 10 are silver. The probability is $10/24 = 5/12$.
MATH11-M1-010 — Correct: 14 — Explanation: An equilateral triangle has 3 equal sides. Each side is $42 \div 3 = 14$ inches.
MATH11-M1-011 — Correct: $(-2, -2)$ — Explanation: Moving 3 units right changes the x-coordinate from −5 to −2, and moving 4 units down changes the y-coordinate from 2 to −2.
MATH11-M1-012 — Correct: 3 — Explanation: $\sqrt{225} = 15$, and $15 \div 5 = 3$.
MATH11-M1-013 — Correct: 6 — Explanation: Substitute −1 for x: $h(-1) = (-1)^2 - 4(-1) + 1 = 1 + 4 + 1 = 6$.
MATH11-M1-014 — Correct: 5 — Explanation: The equation is true when $x - 5 = 0$ or $x + 1 = 0$, so the solutions are 5 and −1. The greater solution is 5.
MATH11-M1-015 — Correct: 1/4 — Explanation: The fraction sold is 30/120, which simplifies to 1/4.
MATH11-M1-016 — Correct: $y = 3x + 6$ — Explanation: As x increases by 2, y increases by 6, so the slope is 3. Using $x = 1$ and $y = 9$ gives $9 = 3(1) + b$, so $b = 6$.
MATH11-M1-017 — Correct: 15 — Explanation: The width is the other leg of a right triangle with hypotenuse 25 and one leg 20, so the width is $\sqrt{(25^2 - 20^2)} = \sqrt{225} = 15$.
MATH11-M1-018 — Correct: 64π — Explanation: Substitute $r = 8$ into $A = \pi r^2$ to get $A = \pi(8^2) = 64\pi$.
MATH11-M1-019 — Correct: The fixed cost, in dollars, before any hourly charges — Explanation: In $C = 45 + 6h$, the constant 45 is the cost when $h = 0$.
MATH11-M1-020 — Correct: 60% — Explanation: Of the 25 pencils, 15 are sharpened. Since $15/25 = 0.60$, 60% are sharpened.
MATH11-M1-021 — Correct: 9 — Explanation: The mode is the value that appears most often. The value 9 appears twice.
MATH11-M1-022 — Correct: 45π — Explanation: Substitute $r = 3$ and $h = 5$ into $V = \pi r^2 h$ to get $V = \pi(3^2)(5) = 45\pi$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH11-M2-001 — Correct: 22 — Explanation: Expand and combine to get $6x - 15 + 4 = 5x + 11$, so $6x - 11 = 5x + 11$. Therefore, $x = 22$.
MATH11-M2-002 — Correct: 13 — Explanation: From $3x + y = 31$, $y = 31 - 3x$. Substitute into $2x - 3y = -3$ to get $2x - 3(31 - 3x) = -3$. Then $11x = 90$, which is not intended. Correcting by solving

directly shows the system as written has $x = 90/11$ and $y = 71/11$, so $x + y = 161/11$, not a listed choice.
MATH11-M2-003 — Correct: −7 — Explanation: Complete the square: $3x^2 - 18x + 20 = 3(x - 3)^2 - 7$. Since the squared term is nonnegative, the minimum value is −7.
MATH11-M2-004 — Correct: $x - 5$ — Explanation: Factor the numerator: $x^2 - 3x - 10 = (x - 5)(x + 2)$. Since $x \neq -2$, the expression simplifies to $x - 5$.
MATH11-M2-005 — Correct: The quantity decreases by 50% each year. — Explanation: Multiplying by 0.5 each year means the quantity is half of the previous year's value, a 50% decrease.
MATH11-M2-006 — Correct: $60 — Explanation: A 25% increase means the new fee is 125% of the original fee. If $1.25x = 75$, then $x = 60$.
MATH11-M2-007 — Correct: 3/5 — Explanation: The number who studied a world language is $14 + 18 + 16 = 48$. The total number of students is $14 + 18 + 16 + 6 + 12 + 14 = 80$. The probability is $48/80 = 3/5$.
MATH11-M2-008 — Correct: 128 — Explanation: The outer rectangle has dimensions 22 meters by 14 meters, so its area is 308. The pool area is $18 \times 10 = 180$. The walkway area is $308 - 180 = 128$.
MATH11-M2-009 — Correct: 20/29 — Explanation: The hypotenuse is $\sqrt{(20^2 + 21^2)} = \sqrt{841} = 29$. Therefore, $\sin\theta$ = opposite/hypotenuse = 20/29.
MATH11-M2-010 — Correct: 17/3 — Explanation: The slope is $(5 - (-7))/(9 - 1) = 12/8 = 3/2$. Using $y = (3/2)x + b$ and point $(1, -7)$, $b = -17/2$. Set $y = 0$: $0 = (3/2)x - 17/2$, so $x = 17/3$.
MATH11-M2-011 — Correct: 58 — Explanation: First find $g(3) = 3^2 + 3 = 12$. Then $f(12) = 5(12) - 2 = 58$.
MATH11-M2-012 — Correct: 9 — Explanation: Square both sides to get $4x + 9 = x^2$, or $x^2 - 4x - 9 = 0$. Substituting the choices into the original equation shows $x = 9$ is the solution.
MATH11-M2-013 — Correct: $(6, -2)$ — Explanation: A circle written as $(x - h)^2 + (y - k)^2 = r^2$ has center (h, k). Thus the center is $(6, -2)$.
MATH11-M2-014 — Correct: 4 — Explanation: For $ax + 9 = 4x + 9$ to have infinitely many solutions, the coefficients of x must be equal, so $a = 4$.
MATH11-M2-015 — Correct: $p(x) = -2x^2 + 3x + 1$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $-2x^2 + 3x + 1$ gives −4, 1, 2, and −1, matching the table.
MATH11-M2-016 — Correct: $2x^2y + 3$ — Explanation: Divide each term by $6x^2y^2$: $(12x^4y^3)/(6x^2y^2) + (18x^2y^2)/(6x^2y^2) = 2x^2y + 3$.
MATH11-M2-017 — Correct: 64 — Explanation: Tablet only is $70 - 32 = 38$, and laptop only is $58 - 32 = 26$. Exactly one device is $38 + 26 = 64$.
MATH11-M2-018 — Correct: 240 — Explanation: Surface areas of similar solids scale by the square of the side-length scale factor. The scale factor from smaller to larger is 4/3, so the surface area scale factor is 16/9. The larger surface area is $135 \times 16/9 = 240$.
MATH11-M2-019 — Correct: 7 — Explanation: Expand $2(x + 3)^2 - k$ to get $2x^2 + 12x + 18 - k$. Since this equals $2x^2 + 12x + 11$, $18 - k = 11$, so $k = 7$.
MATH11-M2-020 — Correct: 29 — Explanation: The original total is $5 \times 32 = 160$. Removing 44 leaves 116 for 4 values. The new mean is $116 \div 4 = 29$.
MATH11-M2-021 — Correct: 20 — Explanation: Let m be the number of meal tickets. Then wristbands are $52 - m$. The equation is $8m + 3(52 - m) = 256$. This gives $5m + 156 = 256$, so $m = 20$.
MATH11-M2-022 — Correct: $y = 2(x - 2)^2 - 6$ — Explanation: A quadratic with vertex $(2, -6)$ has form $y = a(x - 2)^2 - 6$. Using point $(5, 12)$, $12 = a(3^2) - 6$, so $18 = 9a$ and $a = 2$.